

CSCE 633: Machine Learning

Perceptron: Activation Functions

Texas A&M University
Bobak Mortazavi

Perceptron: Recap

- As we saw earlier, a perceptron can be written as:

$$h(x) = \text{sign} \left(\sum_{i=1}^d w_i x_i + w_0 \right)$$

or

$$h(x) = \text{sign} (\mathbf{w}^T \mathbf{x})$$

where $\mathbf{w} = [w_0, w_1, \dots, w_d]^T$ and $\mathbf{x} = [1, x_1, \dots, x_d]^T$

- Note: the bias term w_0 lumped with coefficients w_i for $i \in \{1, 2, \dots, d\}$ and added a corresponding dimension with value 1 to the input feature vector $\mathbf{x}$.
- We stack perceptrons to achieve Multi Layer Perceptrons (MLP).

Perceptron: Activation and Pre-activation

- Single layer perceptron:

$$h(x) = \text{sign}(\mathbf{w}^T \mathbf{x})$$

- Terminology:
 - In the formulation above, the function $\text{sign}()$ is called the *activation function*.
 - The quantity passed to the activation function is called *pre-activations*.
- In MLPs, a hidden unit at layer l is:
 - $z_l = \phi_l(b_l + W_l z_{l-1})$
 - activation: ϕ_l
 - Pre-activation: $b_l + W_l z_{l-1}$

Need for Nonlinearity

- We are free to use any activation function.
- If all activation functions are linear, $\phi_l(a) = c_l a$: (dropping bias terms)

$$f(x; \theta) = W_l c_l (W_{l-1} c_{l-1} (\dots (W_1 x) \dots)) = \mathbf{W}x$$

which is equivalent to a regular linear model.

- Therefore, important to introduce nonlinearity.
- In the perceptron formulation we've seen, $\text{sign}()$ is a nonlinear function. Why?
- We will learn about several activation functions and how they are different in the future.

Conclusion

- In a perceptron, an activation function is applied to pre-activations.
- If all activation functions are linear, the whole MLP collapses to a linear function